GAMEPULSE MARKET INTELLIGENCE

Steam Games Market Analysis Using PostgreSQL, Python, Statistical Analysis and Power BI

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Technology Stack:

PostgreSQL | SQL | Python | Pandas | NumPy | Matplotlib | SciPy | Power BI | Git | GitHub

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ABSTRACT

The gaming industry has experienced rapid growth over the last decade, resulting in the release of thousands of games across digital platforms. Steam is one of the world's largest digital game distribution platforms, hosting tens of thousands of games from developers and publishers worldwide.

The objective of this project is to analyze the Steam Games marketplace and uncover business insights related to pricing, player satisfaction, discounts, ratings, publisher performance, and market growth trends. The project follows a complete Data Analytics workflow consisting of data extraction, transformation, loading into PostgreSQL, SQL-based business analysis, statistical analysis using Python, and visualization through Power BI dashboards.

The analysis was conducted on a dataset containing 71,649 Steam games. Findings reveal that low-priced games dominate the marketplace, game releases increased significantly after 2015, discounts are relatively uncommon, and pricing has little influence on player ratings. The project demonstrates practical implementation of ETL processes, SQL analytics, statistical techniques, and business intelligence reporting.

OBJECTIVE

The primary objectives of this project are:

• Analyze the Steam Games marketplace using data analytics techniques.

• Understand pricing patterns and their impact on player satisfaction.

• Examine game rating trends and review behavior.

• Evaluate the effectiveness of discount strategies.

• Identify top-performing publishers and games.

• Build an end-to-end analytics solution using industry-standard tools.

• Develop interactive dashboards for business decision-making.

DATASET DESCRIPTION

Dataset Name:

Steam Games Dataset

Total Records:

71,649 Games

Dataset Source:

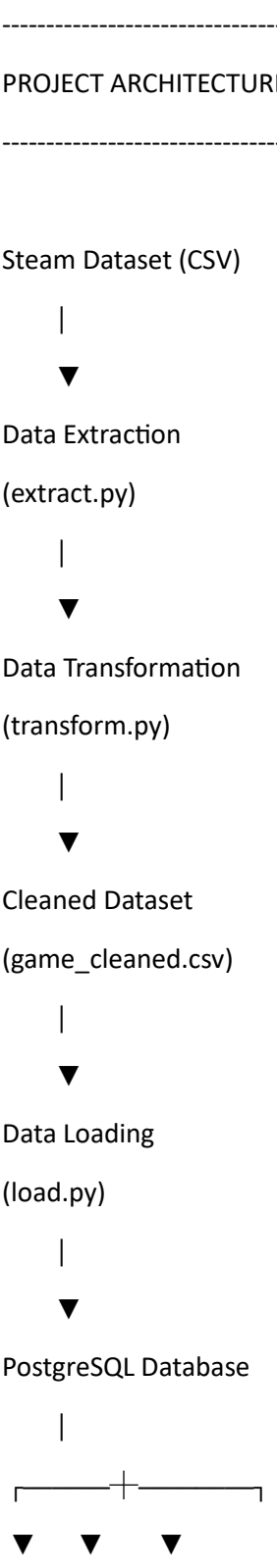
Steam Marketplace Dataset

Key Attributes:

- Game Title
- Release Date
- Publisher
- Developer
- Original Price
- Discount Percentage
- Overall Rating
- Recent Rating
- Review Count
- Platform Information

The dataset provides detailed information regarding game performance, pricing structures, ratings, discounts, and publisher information, making it suitable for business intelligence and market analysis.

PROJECT ARCHITECTURE



SQL Python Power BI

Analysis Stats Dashboard

METHODOLOGY

The project follows a structured Data Analytics lifecycle:

1. Data Collection
2. Data Cleaning
3. Data Transformation
4. Database Loading
5. SQL Business Analysis
6. Statistical Analysis
7. Dashboard Development
8. Insight Generation

Each stage contributes to transforming raw gaming data into actionable business intelligence.

EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis was conducted using Python libraries such as Pandas and Matplotlib to understand the structure, quality, and distribution of the dataset.

The following analyses were performed:

• Dataset Shape Analysis
• Missing Value Analysis

• Duplicate Record Detection

• Data Type Verification

• Price Distribution Analysis

• Rating Distribution Analysis

• Review Count Distribution

• Publisher Analysis

• Release Year Trend Analysis

• Discount Distribution Analysis

Key Observations:

• The dataset contains over 71,000 games.

• Most Steam games are low-priced titles.

• Ratings are concentrated in higher ranges.

• Review counts vary significantly across games.

• Game releases accelerated rapidly after 2015.

• Only a small percentage of games receive discounts.

EDA helped identify patterns and prepare the data for further analysis.

DATA CLEANING PROCESS

Data quality improvement was performed using Python.

Cleaning Steps:

1. Duplicate Removal

- Removed duplicate records.

2. Missing Value Handling

- Addressed missing values where necessary.

3. Data Type Conversion

- Converted columns into appropriate formats.

4. Standardization

- Standardized column names and formatting.

5. Feature Engineering

- Extracted release year.
- Created pricing categories.
- Created rating categories.
- Created discount categories.

Generated Categories:

Price Categories:

• Cheap
• Medium
• Premium

Rating Categories:

• High
• Medium
• Low

Discount Categories:

• No Discount
• Moderate Discount
• Heavy Discount

Output:

game_cleaned.csv

The cleaned dataset was used for SQL analysis and dashboard creation.

POSTGRESQL IMPLEMENTATION

After cleaning, the dataset was loaded into PostgreSQL for business analysis.

Database Activities:

- Database Creation
- Table Creation
- Data Import
- Query Execution
- KPI Calculation

SQL Files Used:

01_schema.sql

02_business_analysis.sql

Benefits of PostgreSQL:

- Efficient querying
- Data integrity
- Scalability
- Advanced aggregation capabilities

SQL BUSINESS ANALYSIS

Multiple SQL queries were written to answer business questions and generate actionable insights.

Analysis Areas:

1. Top Reviewed Games
2. Publisher Performance
3. Price Category Analysis
4. Rating Category Analysis
5. Discount Analysis
6. Release Trend Analysis
7. Market Segmentation

Major SQL Findings:

• Counter-Strike is the most reviewed game on Steam.
• PUBG and Dota 2 rank among the top reviewed titles.
• Cheap games dominate the marketplace.
• Ratings remain consistently high across categories.
• Premium games receive larger average discounts.
• Steam's game release volume increased substantially after 2015.

The SQL layer served as the foundation for business reporting and dashboard metrics.

STATISTICAL ANALYSIS

Statistical analysis was conducted using Python, NumPy, Pandas, and SciPy.

Techniques Used:

- Descriptive Statistics
- Correlation Analysis
- Distribution Analysis
- Comparative Analysis
- Trend Analysis

Key Metrics:

Average Rating:

84.95%

Average Recent Rating:

76.74%

Average Review Count:

15K

Average Price:

$4.64

Median Price:

$2.79

Price-Rating Correlation:

-0.12

Interpretation:

The correlation coefficient of -0.12 indicates a weak negative relationship between price and rating. This suggests that expensive games do not necessarily receive higher player ratings.

Statistical analysis further revealed that player satisfaction remains relatively stable across different pricing and discount categories.

POWER BI DASHBOARD ANALYSIS

The final business intelligence solution consists of three interactive Power BI dashboard pages.

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PAGE 1: STEAM MARKET OVERVIEW

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Purpose:

Provide an executive summary of the Steam marketplace.

KPIs:

- Total Games: 71,649
- Average Price: $4.64
- Median Price: $2.79
- Average Rating: 84.95%
- Average Recent Rating: 76.74%
- Unreleased Games: 10,305

Visualizations:

- Steam Games Released by Year
- Top 10 Most Reviewed Games
- Game Distribution by Price Category
- Rating Distribution

Key Findings:

• Steam experienced significant growth after 2015.

• Cheap games account for the majority of listings.

• Ratings remain strong across the platform.

• Thousands of games remain unreleased.

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PAGE 2: PLAYER SATISFACTION & PRICING ANALYSIS

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Purpose:

Analyze player satisfaction and pricing relationships.

KPIs:

• Average Rating: 84.95%

• Average Recent Rating: 76.74%

• Average Review Count: 15K

• Price-Rating Correlation: -0.12

Visualizations:

• Average Rating by Price Category

• Ratings by Discount Category

• Price vs Rating Scatter Plot

• Review Count vs Rating Scatter Plot

Key Findings:

• Cheap games receive the highest average ratings.

• Premium games receive lower average ratings.

• Pricing has limited influence on player satisfaction.

• Games with larger review counts generally maintain strong ratings.

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PAGE 3: DISCOUNT ANALYSIS & BUSINESS INSIGHTS

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Purpose:

Evaluate discount effectiveness and publisher performance.

KPIs:

• Average Discount: 2.09%

• Discounted Games: 3K

• Discounted Games Percentage: 3.71%

• Average Price: $4.64

Visualizations:

• Distribution of Games by Discount Category

• Average Rating by Discount Category

• Average Discount by Price Category

• Top Publishers by Average Rating

Key Findings:

• Only 3.71% of games are discounted.

• Premium games receive the highest discounts.

• Discounts have minimal influence on ratings.

• Publisher performance varies significantly.

FINDINGS

The analysis produced several important findings:

• Steam hosts more than 71,000 games.

• Average game rating remains high at 84.95%.

• Cheap games receive the highest player ratings.

• Price has little impact on player satisfaction.

• Only 3.71% of games are discounted.

• Premium games receive larger discounts.

• Steam releases increased dramatically after 2015.

• A small number of publishers contribute significantly to marketplace activity.

RECOMMENDATIONS

For Game Publishers:

- Focus on quality rather than premium pricing.
- Monitor player feedback continuously.
- Use discounts strategically instead of frequently.

For Developers:

- Prioritize gameplay quality and user experience.
- Build strong communities to increase review engagement.

For Business Teams:

- Use rating and review metrics to assess product success.
- Analyze market trends before pricing new releases.

For Investors:

- Track high-performing publishers and rapidly growing segments.
- Monitor release trends to identify future opportunities.

CONCLUSION

GamePulse Market Intelligence successfully demonstrates the implementation of a complete end-to-end Data Analytics solution using PostgreSQL, Python, Statistical Analysis, and Power BI.

The project transformed raw Steam marketplace data into meaningful business insights through ETL processing, SQL analysis, statistical modeling, and interactive dashboards.

Key findings show that game quality and player experience contribute more strongly to ratings than pricing. Discounts are relatively uncommon and have limited impact on player satisfaction. The Steam marketplace continues to expand rapidly, presenting opportunities for developers, publishers, and investors.

This project showcases practical skills in data cleaning, database management, SQL analytics, statistical analysis, dashboard development, and business intelligence reporting, making it a strong portfolio project for Data Analyst roles.

END OF REPORT
